

CS 683 : Generative AI

Course Code : CS 683

Course Name : Generative AI

L-T-P-C : 3-0-1-4

Intended for : Elective for B. Tech. Data Science and Engineering, Discipline elective for B. Tech. Computer Science and Engineering, Electrical Engineering, Free elective for other B. Tech. disciplines, Elective for M.Tech (CSP/Intelligent Systems/CSE) and PhD

Prerequisite : CS 671 - Deep Learning and It's Applications

Mutual Exclusion:

Approval: 54th BoA

Course Contents

- **Introduction:** Introduction to Generative Models- GenAI, Basics of Deep Learning ,Convolutional Neural Networks, Convolutional Neural Networks -II, Recurrent Neural Networks & LSTM. (4 Hours)
- **Variational Autoencoders:** Deep Dive Into Variational AutoEncoder, Variants of Variational AutoEncoders. (3 Hours)
- **Generative Adversarial Networks:** Theory ofGAN, Variants ofGAN, GAN architectures, Various GAN losses and Convergence analysis ofGAN, Conditional GANS:pix2pix, cycleGAN, Domain adaptation frameworks. (7 Hours)
- **Flow Based Generative Models:** Normalizing Flow Basics, Variational Normalizing Flow and Sampling Basics. (2 Hours)
- **Diffusion Models:** Introduction to Diffusion Models, Deepdive into Diffusion Models -1, Score Based Diffusion Models, Discrete Latent Variable Models, Diffusion Models for Discrete Data. (7 Hours)
- **Hybrid Generative Models:** V AE-GAN, Diffusion guided Style GAN. (2 Hours)
- **Evaluation of Generative Models:** Discussion ofthe scores like FID, Inception Score. (1 Hour)
- **Transformer Based Generative Models:** Introduction to Transformers, Variants of Transformers, BERT Models, GPT Models, Chat GPT, Video and Image generation via. Transformers: VIT, VVIT, CLIP, GLIDE, DALE, Text Generation, Text to Text and Text to Image, Text to Video. (8 Hours)
- **Deep Graph Generative Models:** Graph Neural Networks- Model, Design, Deep Generative Models for Graph Generation and Graph Transformers. (8 Hours)

Laboratory/practical/tutorial Modules:

- Lab to be conducted on a 2-hour slot every week. It will be conducted in tandem with the theory course so the topics for problems given in the lab are already initiated in the theory class.
- Programming Assignments to Implementing state of art Diffusion Models, GANs,Transformers and Graph Neural Networks.
- Tools- Pytorch or Tensorflow



Textbooks:

1. Alger fraley, **The Artificial Intelligence and Generative AI Bible:** [5 in 1] The Most Updated and Complete Guide I From Understanding the Basics to Delving into GANs, NLP, Prompts, Deep Learning, and Ethics of AI, 2024
2. David Foster, **Generative Deep Learning**, 2nd Edition, O'Reilly Media, Inc.

References:

1. Jakub M. Tomczak, **Deep Generative Modeling**.
2. Keita Broadwater and Namid Stillman, **Graph Neural Networks in Action**.